Assignment 7: 1. PL/SQL Code for finding factorials of a given number

SQL> @E:\SST\Asg_7_1.sql

Code:

set serveroutput on // SET SERVEROUTPUT ON is used to ***redirect messages*** in the ***DBMS_OUTPUT message buffer*** to ***standard output***,

```
declare
n number;
i number;
f number : = 1;
begin
        n : = &x;
        for i in 1..n
        loop
        f := f * i;
        end loop;
        dbms_output.put_line('Factorial of '||n||' is ' || f);
end;
/
```

/ Forward Slash (/): A forward slash is used to ***execute the SQL statement*** or ***PL/SQL block*** that is ***currently in the buffer***.

Output :-

SQL> @E:\SST\Asg_7_1.sql

Enter value for x: 5

old 6: n:=&5;

new 6: n:=5;

Factorial of 5 is 120.

PL/SQL procedure successfully completed.

Assignment 7: 2. PL/SQL code for calculating finding the *sum of N numbers*.

SQL> @E:\SST\Asg_7_2.sql

Code:

```
set serveroutput on
declare
n number;
i number;
s number : = 0;
begin
        n : = &x;
        for i in 1..n
        loop
        s := s + i;
        end loop;
        dbms_output.put_line('Sum of first '||n||' numbers is ' || s);
end;
/
```

Output :-

SQL> @E:\SST\Asg_7_2.sql

Enter value for x: 10

old 6: n:=&x;

new 6: n:=10;

Sum of first 10 numbers is 55.

PL/SQL procedure successfully completed.

Assignment 7: 3. PL/SQL code for finding whether a given year is leap year or not.

SQL> @E:\SST\Asg_7_3.sql

Code:

```
set serveroutput on
declare
y number;
begin
        y=&n;
        if (mod (y,400) = 0 ) then
        dbms_output.put_line('Leap Year');
        elsif  ( (mod (y,4) = 0 ) and (mod (y,100) ! = 0 ) )
        then
        dbms_output.put_line('Leap Year');
        else
        dbms_output.put_line('Not a Leap Year');
        end if;
end;
/
```

Output :-

SQL> @E:\SST\Asg_7_3.sql

Enter value for n: 2004

old 4: y:=&n;

new 4: y:= 2004;

Leap Year

PL/SQL procedure successfully completed.

Assignment 7: 4. PL/SQL code for finding maximum of three numbers. (Input will be given by the user).

SQL> @E:\SST\Asg_7_4.sql

Code:

```
 set serveroutput on
declare
a number;
b number;
c number ;
begin
        a := &a;
        b := &b;
        c := &c;
        if (a>b and a>c)
        then
                dbms_output.put_line (a || ‘ is the maximum’);
        elsif (b>c) then
                 dbms_output.put_line (b || ‘ is the maximum’);
        else
                dbms_output.put_line (c || ‘ is the maximum’);
        end if;
end;
/
```

Output :-

```
SQL> @E:\SST\Asg_7_4.sql
Enter value for a: 7
old 6:       a:=&a;
```

new 6: a:= 7;

Enter value for b: 21

old 7: b:=&b;

new 7: b:= 21;

Enter value for c: 10

old 8: c:=&c;

new 8: c:= 10;

b is the maximum.

PL/SQL procedure successfully completed.

Assignment 7: 5. PL/SQL code for calculate the area of a circle for a value of radius varying from 6 to 10.

Store the *radius and corresponding values of calculated area* in an empty table named Areas, consisting of two columns Radius and Area.

SQL> @E:\SST\Asg_7_5.sql

Code:

```
 set serveroutput on
drop table Areas;
create table Areas (radius number (5,3), area number (10,3));
declare
        r number;
        pi constant  number (8,2 ) = 3.14;
        area  number (10, 2) ;
begin
```

```
        for r in 6..10
        loop
                area := pi * power(r, 2);
                insert into Areas values (r, area);
        end loop;
end;
/
```

Output :-

SQL> @E:\SST\Asg_7_5.sql

Table dropped.

Table created.

SQL> select * from areas;

Radius	Area
6	113.04
7	153.86
8	200.96
9	254.34
10	314

Assignment 7: 6. PL/SQL code that will accept a client_no from the user and adds the amount of Rs. 1000 to bal_due column, for the client who is having a minimum balance of Rs. 6000. The process is fire on client_master.

Column No	Column Name	Data Type	Size	Attributes
1	Client_no	Varchar2	6	Primary key, first letter must start with 'C'
2	Name	Varchar2	30	Not NULL
3	Address1	Varchar2	30	
4	Address2	Varchar2	30	
5	City	Varchar2	15	
6	State	Varchar2	15	
7	Pincode	Number	6	
8	Balance_due	Number	10,2	

Col-1	Col-2	Col-3	Col-4	Col-5	Col-6	Col-7	Col-8
C001	Ivan Bayross	P-76	Worli	Bombay	Maharastra	400054	**15000**
C002	VandanaSatiwal	128	Adams Street	Madras	TamilNadu	780001	0
C003	PramadaJaguste	157	Gopalpur	Kolkata	West Bengal	700058	**5000**
C004	Basu Navindgi	A/12	Nariman	Bombay	Maharastra	400056	0
C005	Ravi Sreedharan	B/34	Rajnagar	Delhi	Delhi	100001	**2000**
C006	Rukmini	Q-12	Bandra	Bombay	Maharastra	400050	0

SQL> @E:\SST\Asg_7_6.sql

Code:

```
 set serveroutput on
declare
        cli_no varchar2(6) := '&client_no';
        t_c_no  number(10,2);  //storing the balance_due value
begin
        select  balance_due into t_c_no from client_master where
        client_no = cli_no;
        if (t_c_no >=6000)
        then
                t_c_no := t_c_no+1000;
                update client_master set balance_due= t_c_no where
                client_no = cli_no;
        else
        dbms_output.put_line (' The balance is below 6000.');
        end if;
end;
/
```

Output :-

```
SQL> @E:\SST\Asg_7_6.sql
Enter value for client_no : C001
old 2:       cli_no varchar (2, 6) := '&client_no';
```

new 2: cli_no varchar (2, 6) := ‘C001’;

PL/SQL procedure successfully completed.

SQL> select client_no, balance_due from client_master;

CLIENT_NO	BALANCE_DUE
C001	17000
C002	0
C003	5000
C004	0
C005	2000
C006	0

Assignment 8: 1.a Create a table whose structure will be as follows:

Table Name: Prime_Entry

Column Name	Data Type	Attributes
Num_id	Number(3)	Primary Key
Prime_num	Number(3)	Not Null

SQL> @E:\SST\Asg_8\Asg_8_1.sql

Code:

```
set serveroutput on;
create table prime_entry(Num_id number(3) primary key, prime_num number(3) not null);
create sequence seq
start with 1
increment by 1
/
```

Output :-

SQL> @E:\SST\Asg_8\Asg_8_1.sql

Table Created.

Sequence Created.

Assignment 8: 1.b Write a PL/SQL block of code that will take a ***number from user*** and test whether the ***number is prime or not***. If the ***number is prime***, then enter into above table by generating ***NUMID automatically***.

SQL> @E:\SST\Asg_8\Asg_8_2.sql

Code:

```
set serveroutput on
declare
        num number;
        j number;
        i number;
        n number;
```

```
flag number;
g number;
begin
        num:=&n;
        n:=TRUNC(num/2);
        for i in 2..n
        loop
        if(mod(num,i)=0) then
                flag:=1;
                exit;
        else
                flag:=0;
        end if;
end loop;
dbms_output.put_line('-----------------');
if(flag=1) then
        dbms_output.put_line(num || ' is not prime  ');
else
        select seq.nextval into g from dual;
        insert into prime_entry values(g, num);
end if;
end;
/
```

Output :-

```
SQL> @E:\SST\Asg_8\Asg_8_2.sql
Enter value for n: 5
old 9:      num:=&n;
new 9:      num:=5;
-------------------------------
PL/SQL procedure successfully completed.
```

SQL> @E:\SST\Asg_8\Asg_8_2.sql

Enter value for n: 7

old 9: num:=&n;

new 9: num:=7;

PL/SQL procedure successfully completed.

SQL>select * from prime_entry;

NUM_ID	PRIME_NUM
1	5
2	7

Assignment 8: 1.c Now add a ***checking*** for ***same prime number entry***. It will show - '**Number already exists in database'** for same ***prime number entry***. Write a ***function to test*** whether ***given number exist or not***.

SQL> @E:\SST\Asg_8\Asg_8_3.sql

Code:

```
set serveroutput on
create or replace function prime_test(id number) return number
is
num number(20);
begin
        select num_id into num from prime_entry where prime_num=id;
        return 1;
        exception
                when no_data_found then
                return 0;
end;
/
```

```
declare
        num number;
        j number;
        i number;
        n number;
        flag number;
        x number;
        begin
                num:=&n;
                n:=TRUNC(num/2);
                for i in 2..n
                loop
                if(mod(num,i)=0) then
                        flag:=1;
                        exit;
                else
                        flag:=0;
                end if;
                end loop;
        dbms_output.put_line('-----------------');
        if(flag=1) then
                dbms_output.put_line(num || ' is not prime number. ');
        else
                x:=prime_test(num);
                if(x=0) then
                insert into prime_entry values(seq.nextval, num);
                else
                dbms_output.put_line('  Already exist in table.  ');
                end if;
        end if;
```

```
end;
/
```

Output :-

```
SQL> @E:\SST\Asg_8\Asg_8_3.sql
Function created;
Enter value for n: 4
old 9:      num:=&n;
new 9:      num:=4;
-------------------------------
4 is not prime number.
PL/SQL procedure successfully completed.
SQL>select * from prime_entry;
   NUM_ID        PRIME_NUM
      1                5
      2                7
```

```
SQL> @E:\SST\Asg_8\Asg_8_3.sql
Function created;
Enter value for n: 5
old 9:      num:=&n;
new 9:      num:=5;
-------------------------------
Already exist in table.
PL/SQL procedure successfully completed.
-------------------------------------
 SQL> @E:\SST\Asg_8\Asg_8_3.sql
Function created;
Enter value for n: 11
old 9:      num:=&n;
```

new 9: num:=11;

PL/SQL procedure successfully completed.

SQL>select * from prime_entry;

NUM_ID	PRIME_NUM
1	5
2	7
3	11

1. **Assignment 8: 2** Create the following table:

Table Name: Acc_details

Column_Name	Data type	Size	Attributes
Acc_no	Varchar2	8	Primary Key
Name	Varchar2	20	Not Null
Address	Varchar2	20	Not Null
DOB	Date		Not Null
Sex	Char	1	Not Null, Values ('M', 'F')
Contact_no	Number	10	Not Null
Last_trans_date	Date		Not Null
Total_amt	Number	12,4	Not Null
Acc_status	Char	1	Not Null, Values ('A', 'I')

Table Name: Transactions_Acc

Column_Name	Data type	Size	Attributes
Transaction_id	Number	8	Primary Key
Acc_no	Number	8	References Acc_details.Acc_no
Deposit_amt	Number	12,4	
Withdraw_amt	Number	12,4	
Mode_trans	Char	5	Not Null
Cheque_no	Number	6	Default 0
Trans_date	Date		Not Null

When a specific account will be deleted from Acc_details, then all the transaction details from Transactions_Acc will be deleted for that account number.

create table Acc_details

```
(
Acc_No varchar2(8) primary key,
Name varchar2(20) not null,
Address varchar2(50) not null,
DOB date not null,
sex char(1) check (sex in ('M', 'F')),
contact_no number(10) not null,
last_trans_date date not null,
Total_cost number(14,2) not null,
Acc_status char(1) not null check(Acc_status in ('A', 'I'))
);

create table Transaction_Acc
(
Transaction_Id number(8) primary key,
Acc_No varchar2(8) references Acc_details on DELETE CASCADE,
Deposit_amt number(12,4),
Withdraw_amt number(12,4),
Mode_trans char(5) not null,
Check_no number(6) default 0,
Trans_date date not null
);
desc Acc_details;
desc Transaction_acc;

insert into Acc_details values('001', 'AMIT', 'BK-256', '12-JAN-2012','M',9836773258,'13-JUN-2012',12000, 'A');
select * from Acc_details;
insert into Transaction_Acc values(002, '001', 11000, 5000, 'A', 101, '12-JUN-2012');
```

```
insert into Transaction_Acc values(003, '001', 12000, 6000, 'B', 102, '12-JUL-2012');
select * from Transaction_Acc;

delete from Acc_details where Acc_no='001';

//1 row deleted.

select * from Acc_details;
//no rows selected.

select * from Transaction_Acc;
//no rows selected.
```

Assignment 9 Solutions:

Asg_9_1.sql

Table Name: Acc_details

Column_Name	Data type	Size	Attributes
Acc_no	Varchar2	8	Primary Key
Name	Varchar2	20	Not Null
Address	Varchar2	20	Not Null
DOB	Date		Not Null
Sex	Char	1	Not Null, Values (’M’, ‘F’)
Contact_no	Number	10	Not Null
Last_trans_date	Date		Not Null
Total_amt	Number	12,4	Not Null
Acc_status	Char	1	Not Null, Values (’A’, ‘I’)

```
create table Acc_details
(
Acc_No varchar2(8) primary key,
Name varchar2(20) not null,
Address varchar2(50) not null,
DOB date not null,
sex char(1) check (sex in ('M', 'F')),
contact_no number(10) not null,
last_trans_date date not null,
Total_cost number(14,2) not null,
Acc_status char(1) not null check(Acc_status in ('A', 'I'))
);
```

```
insert into Acc_details values('001', 'AMIT', 'BK-256', '12-JAN-2012','M',9836773258,'13-JUN-2012',12000, 'A');
insert into Acc_details values('002', 'SUMIT', 'AB-125', '10-FEB-2012','M',9830073258,'13-JAN-2012',1500, 'A');
insert into Acc_details values('003', 'RAMIT', 'BG-350', '25-JAN-2013','M',9877363258,'15-JUL-2012',10000, 'A');
```

```
set serveroutput on
declare
n number(20);
t number(20);
amt number:=500;
begin
        n:=&n;
        update Acc_details set Total_amt=Total_amt-amt where
        Acc_No=n;
        commit;
        savepoint s;
        update Acc_details set Total_amt=Total_amt-amt where
        Acc_No=n;
        select Total_amt into t from Acc_details where Acc_No=n;
        if(t<1000) then
        dbms_output.put_line(' Balance after 2nd Transaction-  ' || t);
        dbms_output.put_line(' Insufficient Balance. ');
        rollback to savepoint s;
        dbms_output.put_line(' Balance after Rollback  ' || t);
        else
        commit;
        select Total_Amt into t from Acc_details where Acc_No=n;
        dbms_output.put_line(' Balance after COMMIT  ' || t);
        end if;
end;
/
SQL> @E:\SST\Asg_9\Asg_9_1.sql
Table created.
```

1 row created.

1 row created.

1 row created.

Enter value for n:= 001

old 6: n:=&n;

new 6: n:=001;

Balance after COMMIT 11000

PL/SQL procedure successfully completed.

SQL>desc Acc_details;

SQL>select * from Acc_details;

```
SQL> select * from Acc_details;

ACC_NO   NAME                ADDRESS                                            DOB       S CONTACT_NO LAST_TRAN  TOTAL_AMT A
-------- ------------------- -------------------------------------------------- --------- - ---------- --------- ---------- -
001      AMIT                BK-256                                             12-JAN-12 M 9836773258 13-JUN-12      11000 A
002      SUMIT               AB-125                                             10-FEB-12 M 9830073258 13-JAN-12       1500 A
003      RAMIT               BG-350                                             25-JAN-13 M 9877363258 15-JUL-12      10000 A

SQL>
```

Asg_9_2.sql

SQL>Select * from Dept;

```
SQL> select * from Dept;

   DEPT_NO DNAME         LOC
---------- -------------- -------------
        10 ACCOUNTING     NEW YORK
        20 RESEARCH       DALLAS
        30 SALES          CHICAGO
        40 OPERATIONS     BOSTON

SQL>
```

CODE:

```
set serveroutput on
declare
        dno number:=&dno;
        loc1 varchar2(10):='&loc';
        begin
                update Dept set loc=loc1 where Dept_no=dno;
                if sql%found then
                dbms_output.put_line('  The updated loc is   ' || loc1);
                end if;
                if sql%notfound then
                dbms_output.put_line('  The updated loc is not found.  ');
                end if;
        end;
/
```

SQL> @E:\SST\Asg_9\Asg_9_2.sql

```
SQL> @C:\DBMSPrograms\Asg_9\Asg_9_2.sql;
Enter value for dno: 20
old   2:        dno number:=&dno;
new   2:        dno number:=20;
Enter value for loc: MUMBAI
old   3:        loc1 varchar2(10):='&loc';
new   3:        loc1 varchar2(10):='MUMBAI';
The updated loc is   MUMBAI

PL/SQL procedure successfully completed.
```

```
003      RAMIT                BG-350                                    25-JAN-13 M 9877363258 15-JUL-12      10000 A

SQL> select * from Dept;

   DEPT_NO DNAME          LOC
---------- -------------- -------------
        10 ACCOUNTING     NEW YORK
        20 RESEARCH       DALLAS
        30 SALES          CHICAGO
        40 OPERATIONS     BOSTON

SQL> @C:\DBMSPrograms\Asg_9\Asg_9_2.sql;
Enter value for dno: 20
old   2:        dno number:=&dno;
new   2:        dno number:=20;
Enter value for loc: MUMBAI
old   3:        loc1 varchar2(10):='&loc';
new   3:        loc1 varchar2(10):='MUMBAI';
The updated loc is   MUMBAI

PL/SQL procedure successfully completed.

SQL> select * from Dept;

   DEPT_NO DNAME          LOC
---------- -------------- -------------
        10 ACCOUNTING     NEW YORK
        20 RESEARCH       MUMBAI
        30 SALES          CHICAGO
        40 OPERATIONS     BOSTON

SQL>
```

Asg_9_3.sql

CODE:

```
set serveroutput on
declare
        ename varchar2(20);
        Eno number:=&Eno;
        begin
                select ename into ename from Emp where Emp_no=Eno;
                dbms_output.put_line('  The Employee name is   ' ||
                ename);
                exception
                when NO_DATA_FOUND then
                dbms_output.put_line('  The Employee is not found for
the given Emp No.  ');
        end;
/
SQL> select * from Emp;
```

```
SQL> select * from Emp;

    EMP_NO ENAME      JOB              MGR HIREDATE         SAL       COMM    DEPT_NO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7839 KING       PRESIDENT            17-NOV-81       5000                    10
      7698 BLAKE      MANAGER          7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER          7839 09-JUN-81       2450                    10
      7566 JONES      MANAGER          7839 02-APR-81       2975                    20
      7788 SCOTT      ANALYST          7566 13-JUL-87       3000                    20
      7902 FORD       ANALYST          7566 03-DEC-81       3000                    20
      7369 SMITH      CLERK            7902 17-DEC-80        800                    20
      7499 ALLEN      SALESMAN         7698 20-FEB-81       1600        300         30
      7521 WARD       SALESMAN         7698 22-FEB-81       1250        500         30
      7654 MARTIN     SALESMAN         7698 28-SEP-81       1250       1400         30
      7844 TURNER     SALESMAN         7698 08-SEP-81       1500          0         30

    EMP_NO ENAME      JOB              MGR HIREDATE         SAL       COMM    DEPT_NO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7876 ADAMS      CLERK            7788 13-JUL-87       1100                    20
      7900 JAMES      CLERK            7698 03-DEC-81        950                    30
      7934 MILLER     CLERK            7782 23-JAN-82       1300                    10
```

SQL> @E:\SST\Asg_9\Asg_9_3.sql

```
SQL> @C:\DBMSPrograms\Asg_9\Asg_9_3.sql;
Enter value for eno: 7934
old   3:        Eno number:=&Eno;
new   3:        Eno number:=7934;
The Employee name is   MILLER
```

Asg_9_4.sql

CODE:

```
set serveroutput on

declare

IN_ERR exception;

Pragma

(The PRAGMA keyword is used to signify that the remainder of the PL/SQL statement is a pragma, or directive, to the compiler. Pragmas are processed at compile time; they do not execute during runtime.) // Comment

exception_init(IN_ERR, -01400); (//Comment: The EXCEPTION_INIT pragma associates a user-defined exception name with an error code.)

begin

insert into Emp values (null,'BLAKE','MANAGER',7839,to_date('1-5-1981','dd-mm-yyyy'),2850, null, 30);

exception

                when IN_ERR then

dbms_output.put_line('  Cannot insert Null values in not Null column.  ');

end;

/

SQL> @E:\SST\Asg_9\Asg_9_4.sql
```

```
SQL> @C:\DBMSPrograms\Asg_9\Asg_9_4.sql;
Cannot insert Null values in not Null column.
```

Asg_9_5.sql

CODE:

```
set serveroutput on

create table Emp_sal_inc(Emp_id number(10), cur_sal
number(10,4), inc_date date);

declare

cursor cur is

select Emp_no, Sal from Emp where Dept_no=(Select Dept_no from
Dept where Dname='RESEARCH');

Emp_id number;

Emp_sal Emp.Sal%type;

begin

                    open cur;

                    if cur%isopen then

                    loop

                    fetch cur into Emp_id, Emp_sal;

                    exit when cur%notfound;

update Emp set Sal=Sal*1.02 where Emp_no=Emp_id;

insert into Emp_sal_inc values(Emp_id, Emp_sal, SYSDATE);

                    end loop;

                    commit;
```

```
    dbms_output.put_line(cur%rowcount);
    else
    dbms_output.put_line('  Cursor not open....');
    end if;
    close cur;
end;
/
```

SQL> select * from Dept;

```
  DEPT_NO DNAME         LOC
--------- ------------- ------------
       10 ACCOUNTING    NEW YORK
       20 RESEARCH      MUMBAI
       30 SALES         CHICAGO
       40 OPERATIONS    BOSTON
```

SQL> @E:\SST\Asg_9\Asg_9_5.sql

SQL> select * from Emp;

```
    EMP_NO ENAME      JOB              MGR HIREDATE         SAL       COMM    DEPT_NO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7839 KING       PRESIDENT            17-NOV-81       5000                    10
      7698 BLAKE      MANAGER          7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER          7839 09-JUN-81       2450                    10
      7566 JONES      MANAGER          7839 02-APR-81     3034.5                    20
      7788 SCOTT      ANALYST          7566 13-JUL-87       3060                    20
      7902 FORD       ANALYST          7566 03-DEC-81       3060                    20
      7369 SMITH      CLERK            7902 17-DEC-80        816                    20
      7499 ALLEN      SALESMAN         7698 20-FEB-81       1600        300         30
      7521 WARD       SALESMAN         7698 22-FEB-81       1250        500         30
      7654 MARTIN     SALESMAN         7698 28-SEP-81       1250       1400         30
      7844 TURNER     SALESMAN         7698 08-SEP-81       1500          0         30

    EMP_NO ENAME      JOB              MGR HIREDATE         SAL       COMM    DEPT_NO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7876 ADAMS      CLERK            7788 13-JUL-87       1122                    20
      7900 JAMES      CLERK            7698 03-DEC-81        950                    30
      7934 MILLER     CLERK            7782 23-JAN-82       1300                    10

14 rows selected.
```

SQL> select * from Emp_sal_inc;

```
SQL> select * from Emp_sal_inc;

    EMP_ID    CUR_SAL INC_DATE
---------- ---------- ---------
      7566       2975 12-MAY-23
      7788       3000 12-MAY-23
```

SQL> @E:\SST\Asg_9\Asg_9_5.sql

```
SQL> @C:\DBMSPrograms\Asg_9\Asg_9_5.sql

Table created.

5

PL/SQL procedure successfully completed.
```

```
set serveroutput on
declare
        cursor add_interest
        is
        select Acc_no, Total_amt from Acc_details where Acc_status='A';
        varaccn Acc_details.Acc_no%type;
        varamt Acc_details.Total_amt%type;
                begin
                open add_interest;
                if add_interest%isopen then
                        loop
                        fetch add_interest into varaccn, varamt;
                        exit when add_interest%notfound;
        update Acc_details set Total_amt=varamt*1.02 where
        Acc_no=varaccn;
        dbms_output.put_line( varaccn || '   is updated  ');
        end loop;
                else
                        dbms_output.put_line('Cursor not opened.  ');
                end if;
                close add_interest;
                commit;
                end;
/
```

```
set serveroutput on
create function Max_id return number
is
        var_id number(4);
        begin
        select max(Transaction_id) into var_id from Transaction_acc;
                if var_id is null then
                var_id:=200;
                else
                var_id:=var_id+1;
                end if;
                return var_id;
                exception
                when no_data_found then
                return var_id;
        end;
/

create procedure Transaction_entry(varaccn in
Acc_details.Acc_no%type, varamt in Acc_details.Total_amt%type)
        is
        vartid Transaction_acc.Transaction_id%type;
        begin
        vartid:=Max_id();
        insert into Transaction_acc values(vartid, varaccn,varamt, 0,
        'CHQ',0,Sysdate);
        dbms_output.put_line('  Data inserted with Id   ' ||vartid);
        end;
```

```
/
declare
        cursor add_interest
        is
        select Acc_no, Total_amt from Acc_details where
        Acc_status='A';
        varaccn Acc_details.Acc_no%type;
        varamt Acc_details.Total_amt%type;
        begin
        open add_interest;
        if add_interest%isopen then
            loop
            fetch add_interest into varaccn, varamt;
            exit when add_interest%notfound;
        update Acc_details set Total_amt=varamt*1.02 where
        Acc_no=varaccn;
        dbms_output.put_line( varaccn || '   is updated  ');
        varamt:=varamt*1.02;
        Transaction_entry(varaccn, varamt);
        end loop;
        else
            dbms_output.put_line('Cursor not opened.  ');
        end if;
        close add_interest;
        commit;
        end;
/
```

```
set serveroutput on
create table Emp_audit
(Emp_no number(4) primary key,
 Dept_no number(4) not null references Dept,
 Status varchar2(8),
 Salary number(8,2) not null,
 Audit_date date not null);
```

```
set serveroutput on
drop trigger trg_sal;
create trigger trg_sal after
update or delete on Emp for each row
  declare
                status varchar2(20);
                begin
                        if updating then
                        status:='UPDATE';
                        end if;
                        if deleting then
                        status:='DELETE';
                        end if;
        insert into Emp_audit values(:Old.Emp_no,
:Old.Dept_no,status, :Old.Sal,SYSDATE);
                end;
/
```

Assg. 10: Solutions

SQL> @C:\DBMSPrograms\Asg_10\Asg_10_A.sql;

```
SQL> select * from Acc_details;

ACC_NO   NAME                 ADDRESS                                            DOB       S CONTACT_NO LAST_TRAN  TOTAL_AMT A
-------- -------------------- -------------------------------------------------- --------- - ---------- --------- ---------- -
001      AMIT                 BK-256                                             12-JAN-12 M 9836773258 13-JUN-12      12000 A
002      SUMIT                AB-125                                             10-FEB-12 M 9830073258 13-JAN-12       1500 A
003      RAMIT                BG-350                                             25-JAN-13 M 9877363258 15-JUL-12      10000 A

SQL> @C:\DBMSPrograms\Asg_10\Asg_10_A
001is updated
002is updated
003is updated

PL/SQL procedure successfully completed.

SQL> select * from Acc_details;

ACC_NO   NAME                 ADDRESS                                            DOB       S CONTACT_NO LAST_TRAN  TOTAL_AMT A
-------- -------------------- -------------------------------------------------- --------- - ---------- --------- ---------- -
001      AMIT                 BK-256                                             12-JAN-12 M 9836773258 13-JUN-12      12240 A
002      SUMIT                AB-125                                             10-FEB-12 M 9830073258 13-JAN-12       1530 A
003      RAMIT                BG-350                                             25-JAN-13 M 9877363258 15-JUL-12      10200 A
```

```
SQL> create table Transaction_Acc
  2  (
  3  Transaction_Id number(8) primary key,
  4  Acc_No varchar2(8) references Acc_details on DELETE CASCADE,
  5  Deposit_amt number(12,4),
  6  Withdraw_amt number(12,4),
  7  Mode_trans char(5) not null,
  8  Check_no number(6) default 0,
  9  Trans_date date not null
 10  );

Table created.

SQL> insert into Transaction_Acc values(002, '001', 11000, 5000, 'A', 101, '12-JUN-2012');

1 row created.

SQL> insert into Transaction_Acc values(003, '001', 12000, 6000, 'B', 102, '12-JUL-2012');

1 row created.
```

```
SQL> insert into Transaction_Acc values(002, '001', 11000, 5000, 'A', 101, '12-JUN-2012');

1 row created.

SQL> insert into Transaction_Acc values(003, '001', 12000, 6000, 'B', 102, '12-JUL-2012');

1 row created.

SQL> select * from Transaction_Acc;

TRANSACTION_ID ACC_NO   DEPOSIT_AMT WITHDRAW_AMT MODE_   CHECK_NO TRANS_DAT
-------------- -------- ----------- ------------ ----- ---------- ---------
             2 001            11000         5000 A            101 12-JUN-12
             3 001            12000         6000 B            102 12-JUL-12
```

SQL> @C:\DBMSPrograms\Asg_10\Asg_10_B.sql;

```
SQL> @C:\DBMSPrograms\Asg_10\Asg_10_B

Function created.


Procedure created.

001   is updated
Data inserted with Id   4
002   is updated
Data inserted with Id   5
003   is updated
Data inserted with Id   6

PL/SQL procedure successfully completed.

SQL> select * from Acc_details;

ACC_NO   NAME                 ADDRESS                                            DOB       S CONTACT_NO LAST_TRAN  TOTAL_AMT A
-------- -------------------- -------------------------------------------------- --------- - ---------- --------- ---------- -
001      AMIT                 BK-256                                             12-JAN-12 M 9836773258 13-JUN-12    12484.8 A
002      SUMIT                AB-125                                             10-FEB-12 M 9830073258 13-JAN-12     1560.6 A
```

SQL> select * from Acc_details;

SQL> select * from Transaction_acc;

```
PL/SQL procedure successfully completed.

SQL> select * from Acc_details;

ACC_NO   NAME                 ADDRESS                                            DOB       S CONTACT_NO LAST_TRAN  TOTAL_AMT A
-------- -------------------- -------------------------------------------------- --------- - ---------- --------- ---------- -
001      AMIT                 BK-256                                             12-JAN-12 M 9836773258 13-JUN-12    12484.8 A
002      SUMIT                AB-125                                             10-FEB-12 M 9830073258 13-JAN-12     1560.6 A
003      RAMIT                BG-350                                             25-JAN-13 M 9877363258 15-JUL-12       10404 A

SQL> select * from Transaction_Acc;

TRANSACTION_ID ACC_NO   DEPOSIT_AMT WITHDRAW_AMT MODE_   CHECK_NO TRANS_DAT
-------------- -------- ----------- ------------ ----- ---------- ---------
             2 001            11000         5000 A            101 12-JUN-12
             3 001            12000         6000 B            102 12-JUL-12
             4 001          12484.8            0 CHQ            0 14-MAY-23
             5 002           1560.6            0 CHQ            0 14-MAY-23
             6 003            10404            0 CHQ            0 14-MAY-23
```

Solu: 10_2.SQL

SQL> select * from Acc_details;

```
SQL Plus

14 rows selected.

SQL> set linesize 300;
SQL> select * from Emp;

    EMP_NO ENAME      JOB              MGR HIREDATE         SAL       COMM    DEPT_NO
---------- ---------- --------- --------- --------- ---------- ---------- ----------
      7839 KING       PRESIDENT           17-NOV-81       5000                    10
      7698 BLAKE      MANAGER        7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER        7839 09-JUN-81       2450                    10
      7566 JONES      MANAGER        7839 02-APR-81     3034.5                    20
      7788 SCOTT      ANALYST        7566 13-JUL-87       3060                    20
      7902 FORD       ANALYST        7566 03-DEC-81       3060                    20
      7369 SMITH      CLERK          7902 17-DEC-80        816                    20
      7499 ALLEN      SALESMAN       7698 20-FEB-81       1600        300         30
      7521 WARD       SALESMAN       7698 22-FEB-81       1250        500         30
      7654 MARTIN     SALESMAN       7698 28-SEP-81       1250       1400         30
      7844 TURNER     SALESMAN       7698 08-SEP-81       1500          0         30

    EMP_NO ENAME      JOB              MGR HIREDATE         SAL       COMM    DEPT_NO
---------- ---------- --------- --------- --------- ---------- ---------- ----------
      7876 ADAMS      CLERK          7788 13-JUL-87       1122                    20
      7900 JAMES      CLERK          7698 03-DEC-81        950                    30
      7934 MILLER     CLERK          7782 23-JAN-82       1300                    10

14 rows selected.

SQL> desc Emp;
 Name
      Null?      Type
 ----------------------------------------------------------------------------------------------------------------------------
```

SQL>@C:\DBMSPrograms\Asg_10\Asg_10_2a.sql;

Table created;

SQL>@C:\DBMSPrograms\Asg_10\Asg_10_2b.sql;

Trigger dropped;

Trigger created;

SQL) update Emp set Sal=2050 where Emp_no=7499;

```
SQL Plus

no rows selected

SQL> update Emp set Sal=2050 where Emp_no=7499;

1 row updated.

SQL> select * from Emp;

    EMP_NO ENAME      JOB              MGR HIREDATE         SAL       COMM    DEPT_NO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7839 KING       PRESIDENT            17-NOV-81       5000                    10
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER         7839 09-JUN-81       2450                    10
```

SQL) select * from Emp_audit;

```
SQL> select * from Emp_audit;

    EMP_NO    DEPT_NO STATUS       SALARY AUDIT_DAT
---------- ---------- -------- ---------- ---------
      7499         30 UPDATE         2000 14-MAY-23

SQL>
```